

SQL JOINS EXPLAINED WITH ONE DATABASE

Employees

<u>Customer_ID</u>	<u>Customer_Name</u>
101	John Smith
102	Sarah Johnson
103	Michael Brown
104	Emma Davis

Departments

<u>Dept_ID</u>	<u>Dept_Name</u>
101	HR
102	IT
104	Finance

INNER JOIN

 Returns only matching records from both tables.

Query

```
SELECT c.Customer_Name, o.Order_ID  
FROM Customers c  
INNER JOIN Orders o  
ON c.Customer_ID = o.Customer_ID;
```

Output

John Smith → 1001

Sarah Johnson → 1002

Explanation

-  Only customers who placed orders
-  Michael Brown removed
-  Emma Davis removed
-  Order 1003 removed (Customer 105 doesn't exist)

 INNER JOIN = Matching Records Only

2 LEFT JOIN

 Returns all customers and matching orders.

Query

```
SELECT c.Customer_Name, o.Order_ID  
FROM Customers c  
LEFT JOIN Orders o  
ON c.Customer_ID = o.Customer_ID;
```

Output

John Smith → 1001

Sarah Johnson → 1002

Michael Brown → NULL

Emma Davis → NULL

Explanation

 All customers are shown

 Customers without orders get NULL

 LEFT JOIN = Keep All Customers

3 RIGHT JOIN

 Returns all orders and matching customers.

Query

```
SELECT c.Customer_Name, o.Order_ID  
FROM Customers c  
RIGHT JOIN Orders o  
ON c.Customer_ID = o.Customer_ID;
```

Output

John Smith → 1001
Sarah Johnson → 1002
NULL → 1003

Explanation

 All orders are shown

 Order 1003 belongs to a customer not found in Customers table

 RIGHT JOIN = Keep All Orders

FULL OUTER JOIN

 Returns all customers and all orders.

Query

```
SELECT c.Customer_Name, o.Order_ID  
  
FROM Customers c  
  
FULL OUTER JOIN Orders o  
  
ON c.Customer_ID = o.Customer_ID;
```

Output

John Smith → 1001

Sarah Johnson → 1002

Michael Brown → NULL

Emma Davis → NULL

NULL → 1003

Explanation

 Includes matching and non-matching records from both tables

 FULL JOIN = Everything

CROSS JOIN

 Returns every possible combination.

Query

```
SELECT c.Customer_Name, o.Order_ID  
  
FROM Customers c  
  
CROSS JOIN Orders o;
```

Output (Sample)

John Smith → 1001
John Smith → 1002
John Smith → 1003
Sarah Johnson → 1001
...

Explanation

 Total Rows = 4 Customers × 3 Orders = 12 Rows

 CROSS JOIN = Every Possible Combination

SELF JOIN

Employees

Employee_ID	Employee_Name	Manager_ID
1	David	NULL
2	Olivia	1
3	Ethan	1
4	Sophia	2

Query

```
SELECT e.Employee_Name AS Employee,  
m.Employee_Name AS Manager  
FROM Employees e  
JOIN Employees m  
ON e.Manager_ID = m.Employee_ID;
```

Output

Olivia → David
Ethan → David
Sophia → Olivia

Explanation

✓ Used to find Employee–Manager relationships

 SELF JOIN = Same Table Joined with Itself



QUICK REVISION

- ◆ INNER JOIN → Matching Records Only
- ◆ LEFT JOIN → All Customers + Matching Orders
- ◆ RIGHT JOIN → All Orders + Matching Customers
- ◆ FULL JOIN → Everything
- ◆ CROSS JOIN → Every Combination
- ◆ SELF JOIN → Same Table with Itself